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Pacific-Basin Finance Journal

Published: 01/12/2025

Publisher's PDF, also known as Version of record

Cyswllt i'r cyhoeddiad / Link to publication

Dyfyniad o'r fersiwn a gyhoeddwyd / Citation for published version (APA):
Wu, H., Gao, Z., Nie, B., & Zhao, B. (2025). Can machines learn Chinese mutual funds? *Pacific-Basin Finance Journal*, *94*.
https://www.sciencedirect.com/science/article/pii/S0927538X25002720

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26. Oct. 2026

Pacific-Basin Finance Journal 94 (2025) 102935



Contents lists available at ScienceDirect

Pacific-Basin Finance Journal

journal homepage: www.elsevier.com/locate/pacfin



Can machines learn Chinese mutual funds?☆

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ARTICLE INFO

JEL Classification:
G11
G17
G23

Keywords:
Machine learning
Chinese mutual fund
Fund selection
Forecast combination

ABSTRACT

This study applies machine learning techniques to predict mutual fund performance and construct high-performing portfolios in the Chinese mutual fund market. Using Random Forest, XGBoost, and LightGBM models, we frame fund selection as a classification problem, estimating each fund's probability of outperforming the market benchmark. Filtered fund characteristics serve as predictors, allowing us to build investable portfolios ranked by predicted outperformance likelihood. Empirical results reveal that machine learning-based portfolios consistently outperform the benchmark, with LightGBM delivering the strongest performance across key metrics. Furthermore, a flexible machine learning combination framework is introduced to improve predictive robustness and portfolio stability. These findings highlight the practical value of machine learning for fund selection and strategy design in complex financial environments.

1. Introduction

The Chinese mutual fund industry has achieved significant headway in the past decade. By the second quarter of 2024, its total open-end fund assets were more than ten times those at the end of 2011 and reached $3.74 trillion. This places China first among emerging markets and fourth globally.[1] According to the *Asset Management Association of China*, as of December 2020, the number of Chinese mutual fund investors had reached nearly 700 million. This figure not only underscores the massive retail investor participation but also indicates that mutual funds have become a cornerstone of long-term financial planning for many Chinese families seeking professionally managed, diversified financial products.

The importance of mutual funds has been further elevated with the introduction of Chinese private pension plans.[2] Under this newly implemented personal pension policy, individuals can allocate their money to mutual funds and pension target funds (a special type of FoF that adjusts the mix of investments in anticipation of the expected retirement date). Such integration of mutual funds within private pension structures reinforces their centrality in shaping individuals' retirement benefits and long-term financial

☆ This article is part of a Special issue entitled: 'ML and Big Data' published in Pacific-Basin Finance Journal.

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[1] Exclude funds of funds (FoFs), seeing the Quarterly Worldwide Mutual Fund Market by the Investment Company Institute (https://www.ici.org/system/files/2024-09/ww_q2_24_public_report_us.xlsx).

[2] It was officially launched on November 25, 2022, as the third pillar in the standard three-pillar pension system (i.e., state, occupational, and private pension provision).

https://doi.org/10.1016/j.pacfin.2025.102935
Received 10 December 2024; Received in revised form 11 September 2025; Accepted 13 September 2025
Available online 16 September 2025
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security. Moreover, mutual funds are key underlying assets for FoFs, making them a primary focus for FoF managers. Consequently, the ability to select Chinese mutual funds ex-ante is of great importance to both retail and institutional investors, driving our research.[3]

Compared with retail investors, fund managers have superior capabilities, such as stock picking and factor timing, to generate market-beating performance (Barras et al., 2022). Nonetheless, the evidence on whether U.S. mutual funds consistently outperform the market, once luck is distinguished from skill, remains mixed. For example, Kosowski et al. (2006) posit that over 10 % of funds can beat the market. However, Fama and French (2010) point out that less than 1 % can achieve significant alpha. Similarly, Barras et al. (2010) find that the proportion of skilled funds dramatically decreases to a negligible level. These puzzling results may imply why passive U.S. mutual funds, including index mutual funds and exchange-traded funds (ETFs), manage comparable or more assets than active ones. However, actively managed mutual funds account for a much greater proportion than passive ones in China. Moreover, mutual funds themselves are part of the institutional market participants, so they are inevitably affected by stock market features. As for the Chinese stock market, its most significant difference from developed stock markets is the investor structure: retail investors dominate the Chinese stock market, whereas institutional investors dominate the developed stock markets (Leippold et al., 2022). Hence, Chinese mutual funds may take advantage of the market inefficiency resulting from the unique investor structure and regulations (e.g., restrictions on short-selling channels for individual investors) to surpass the market (Yi et al., 2018; Cornell et al., 2020), which underlines the feasibility of selecting outperforming funds in China.[4]

To this end, we explore the return predictability of Chinese mutual funds in this study. Previous studies have proposed various fund performance predictors (Jones and Mo, 2021), which lay a solid foundation to address the central problem of how to choose funds ex-ante (Grønborg et al., 2021). As a result, we can sift through thousands of mutual funds to find a set of winning funds by utilizing the joint predictive power of a plethora of predictors. This amounts to extracting efficient information from the information set composed of various variables. However, employing traditional linear regression models for prediction makes it hard to deal with the high-dimensional financial time series, not least to discover the nonlinearities in the relation between fund characteristics and performance and interactions between predictors (Welch and Goyal, 2008; Jones and Mo, 2021). Fortunately, machine learning methods can handle these issues. For instance, traditional linear regression models can be extended under the supervised machine learning framework by imposing regularization on estimated parameters to solve the overfitting issue and achieve dimension reduction. Additionally, tree-based models and neural networks can help solve nonlinearity and interaction issues. Meanwhile, the stock and hedge fund markets have witnessed the success of machine learning prediction techniques (see, e.g., Gu et al., 2020; Wu et al., 2021). Therefore, we apply machine learning techniques to select outperforming funds in China.

Specifically, we develop a structured methodology to predict mutual fund performance and construct outperforming portfolios using machine learning techniques. We formulate fund selection as a classification problem, emphasizing outperformance relative to the fund market benchmark rather than absolute returns. Beginning with sample selection and predictor construction, fund characteristics are refined through statistical and machine learning-based feature selection methods to enhance interpretability and reduce complexity. Three tree-based ensemble models, including Random Forest, XGBoost, and LightGB, are trained and optimized through K-fold cross-validation and applied to classify funds based on their likelihood of outperforming the benchmark index, the Wind Equity Oriented Fund Index. This classification enables the construction of investable portfolios that consistently deliver superior returns, with top-ranked portfolios achieving annualized returns exceeding 15 % and significant risk-adjusted alphas. Among the methods, LightGBM demonstrates the strongest performance, achieving a 16.44 % annualized return and a Sharpe ratio of 0.70. Furthermore, we introduce a flexible machine learning combination framework to enhance portfolio construction, yielding improved portfolio monotonicity and robustness. These results validate the potential of machine learning approaches, particularly tree-based models, in optimizing fund selection and delivering consistent outperformance in the Chinese mutual fund market. This approach is well-suited to this unique market, characterized by high retail investor participation and distinctive regulatory and market features, which introduce both noise and complexity into the data.

In a nutshell, we contribute to two strands of literature. First, we enrich the growing body of research on machine learning applications in financial markets by demonstrating the effectiveness of tree-based ensemble models in predicting mutual fund performance. Unlike prior studies focused on equity or other asset classes (see, e.g., Chinco et al., 2019; Freyberger et al., 2020; Gompers et al., 2020; Gu et al., 2020; Bali et al., 2020; Dong et al., 2022; Leippold et al., 2022; Avramov et al., 2023), we specifically investigate the underexplored domain of mutual funds in the Chinese market, characterized by its distinctive volatility, retail dominance, and regulatory environment. This market-specific focus expands the geographical and contextual scope of existing literature. Second, our work bridges the gap between methodological advancement and practical utility in fund selection. While prior studies often emphasize approaches to predict fund risk-adjusted returns (DeMiguel et al., 2023; Kaniel et al., 2023), we pivot to a classification framework to identify outperforming funds over the fund market benchmark. This framework employed in our study offers the advantage of providing the probability that a fund will outperform the benchmark, allowing FoF fund managers to make more informed and confident decisions. Additionally, by incorporating feature selection and hyperparameter tuning, we address key challenges related to dimensionality and overfitting, providing a robust roadmap for integrating machine learning into fund performance prediction

[3] Given the shift in investor interest from money market funds to equity funds and the close ties between equity funds and the stock market, our study concentrates on Chinese stock funds and equity oriented funds.

[4] As illustrated in Figure A1 of the Online Appendix, the overall Fund Index (the Wind Equity Oriented Fund Index) has consistently outperformed key market indices, reflecting, to some extent, the capabilities of Chinese mutual funds. Furthermore, Figure A2 highlights that within the universe of all funds, many funds can yield superior performance compared to the average fund market. These findings underscore both the feasibility and the necessity of selecting outperforming funds in the Chinese market.

research.

The remainder of the paper is organized as follows: Section 2 discusses related literature. Section 3 describes our data. Section 4 introduces the employed machine learning methods. Section 5 exhibits the main empirical results, followed by conclusions in Section 6.

2. Literature review

2.1. Machine learning in financial markets

There is growing evidence from both academia and industry demonstrating the potential of machine learning for assisting financial institutions and investors to make informed decisions. Drawing from venture capital (VC) principles, the financial investment process can be broadly categorized into three distinct phases: pre-investment, during-investment, and post-investment (Gompers et al., 2020). In each of these stages, machine learning has the potential to offer significant contributions. Particularly, in the pre-investment phase, a crucial step is scrutinizing the investment landscape and identifying the securities for portfolio construction in the subsequent investment stage. This is where return prediction becomes paramount. In the theoretical light of Kelly et al. (2024), the virtue of complexity inherent in return prediction underscores the importance and rationale behind harnessing machine learning. Existing research has utilized various machine learning techniques to predict returns across different financial instruments, including stocks (Chinco et al., 2019; Freyberger et al., 2020; Gompers et al., 2020; Gu et al., 2020; Dong et al., 2022; Leippold et al., 2022; Avramov et al., 2023), bonds (Bali et al., 2020; Bianchi et al., 2021), options (Büchner and Kelly, 2022), and hedge funds (Wu et al., 2021). Our study continues the direction of exploring the application of machine learning in financial markets by focusing on mutual funds.

2.2. Fund performance predictability

Existing studies have documented various fund performance predictors that can be classified into three streams. The first stream is fund-specific characteristics, such as total net assets (Pástor et al., 2015; Zhu, 2018), fund age (Bai et al., 2019; Jones and Mo, 2021), and fund flow volatility (DeMiguel et al., 2023). In addition, Shang and Sutcliffe (2024) find that the ownership structure of fund management companies can influence mutual fund performance. Jiang et al. (2020) find that retail investors in Chinese mutual funds with higher levels of financial literacy tend to achieve better investment outcomes. The second stream is the metrics on the historical performance of funds. For example, the seminal work of Carhart (1997) points out that the one-year four-factor alpha of mutual funds exhibits performance predictability. Additionally, recent studies, including DeMiguel et al. (2023) and Kaniel et al. (2023), highlight the predictive power of performance momentum metrics in forecasting mutual fund returns, such as the t-statistics of factor loadings (betas) from rolling-window time-series regressions of fund returns on the six-factor model (Fama-French five-factor model augmented with a momentum factor), information ratio, and Sharpe ratio. The last stream focuses on holding-based features derived from possessed stocks. For example, Alexander et al. (2007) find that the performance of mutual funds is related to their trade motivations, including value-motivated trading, liquidity-motivated trading, and window-dressing trading. Window dressing is the practice of rebalancing a fund's portfolio before the end of the reporting period by increasing exposure to winning stocks while offloading losing stocks. This behavior has also been documented in the Chinese fund market (Yu et al., 2023).

While these predictors have traditionally been examined using linear regressions or factor-based models, such methods often struggle to capture nonlinear relationships and complex interactions among variables. In contrast, recent advances in machine learning provide flexible, data-driven tools capable of integrating high-dimensional features from all three streams to improve predictive accuracy in fund performance. In this study, we utilize predictors from these three streams in the machine learning framework.[5]

2.3. Fund selection with machine learning

Comparatively speaking, leveraging machine learning for mutual funds is a nascent field of research. To the best of our knowledge, there are only four independent studies that employ machine learning in mutual fund performance prediction. Li and Rossi (2020) use a group of firm characteristics of stock holdings to predict fund performance via the boosting regression tree model, and portfolios built by funds with superior prediction performance can deliver a yearly 2.88 % alpha relative to the Carhart four-factor model. Similarly, DeMiguel et al. (2023) employ multiple machine learning methods, including gradient boosting, Random Forest, and elastic net, to

[5] Our study does not incorporate the full range of known predictors of fund performance, as many predictors identified in prior studies rely on proprietary or difficult-to-access datasets (e.g., fund manager characteristics, shareholder letter sentiment, or distribution channels), which are beyond the scope of our current data environment. Additionally, the literature on fund performance is vast and continually expanding, making exhaustive coverage impractical within a single study. Importantly, our machine learning framework is flexible. It could be extended to integrate other classical or emerging predictors such as turnover (Pástor et al., 2017), peer benchmarks (Evans et al., 2024), social responsibility (El Ghoul and Karoui, 2022), fund managers' family descent and facial attractiveness (Chuprinin and Sosyura, 2018; Bai and Tian, 2023), distribution channels (Bergstresser et al., 2008), sentiment in shareholder letters (Hillert et al., 2024), industry concentration (Kacperczyk et al., 2005), active share (Cremers and Petajisto, 2009), the R-squared from multifactor regressions (Amihud and Goyenko, 2013), maximum drawdowns (Riley and Yan, 2022), offshore exposure concentration (Bai et al., 2022), alpha adjusted for beta-matched passive portfolios (Irvine et al., 2024), an derivative use (Kaniel and Wang, 2025).

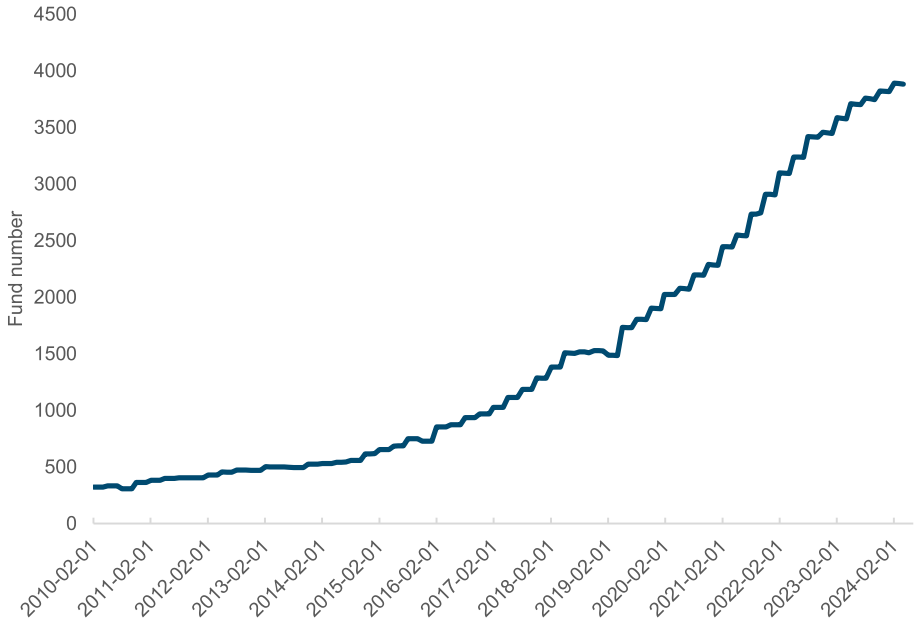


Fig. 1. Number of Funds.
Fig. 1 shows the number of funds according to our criteria from January 31, 2010, to March 31, 2024.

Table 1
Mutual Fund Characteristics.

Type	Variable	Definition
Fundamentals	TNA	Total net assets (RMB million)
	Fund_Age	Number of days since the inception date
	Vol_Flows	The standard deviation of monthly flows in the past 12 months
Performance Momentum	Const_T	Constant from the regression on six-factor model for previous T trading days
	R_Squared_T	R squared
	t_Const_T	Constant t-statistics
	t_MKT_T	Market beta t-statistics
	t_SMB_T	Size beta t-statistics
	t_HML_T	Value beta t-statistics
	t_RMW_T	Profitability beta t-statistics
	t_CMA_T	Investment beta t-statistics
	t_UMD_T	Momentum beta t-statistics
	IR_P6M	Past 6-month Information ratio: (Fund return - Benchmark return)/Tracking error
	IR_P1Y	Past 1-year Information ratio: (Fund return - Benchmark return)/Tracking error
	IR_P3Y	Past 3-year Information ratio: (Fund return - Benchmark return)/Tracking error
	SR_P6M	Past 6-month Sharpe ratio
	SR_P1Y	Past 1-year Sharpe ratio
	SR_P3Y	Past 3-year Sharpe ratio
Holder Structure	Institute	Institutional shareholding ratio
	Worker	Employee shareholding ratio
Trading Characteristics	TS_Buy_Mean	Value-motivated trading motivation for buying, the average of the proportion of buy amounts for stock relative to the fund's TNA at the beginning of the period
	TS_Buy_Std	Value-motivated trading motivation for buying, the variance of the proportion of buy amounts for stock relative to the fund's TNA at the beginning of the period
	SF	Value-motivated trading motivation for selling
	SBRG_Mean	Window-dressing trading
	TRADE_INCs	Stock spread income divided by TNA at the beginning of the report period

This table lists the 61 monthly fund characteristics. "_T" stands for 90, 120, 180, 250, and 750 trading days, resulting in 45 "_T" fund characteristics in total.

choose mutual funds based on fund-specific features. Besides, DeMiguel et al. (2023) prove that long-only portfolios with fund winners selected by nonlinear machine learning methods (gradient boosting and Random Forest) can yield better out-of-sample performance. Moreover, Xia (2021) utilizes penalized linear regression (LASSO, elastic net, and group LASSO) and dimension reduction models (principal component regression and partial least squares) for exploring mutual fund performance from the perspective of retail

investors. Further, Kaniel et al. (2023) conjecture that only fund momentum and flow can predict future risk-adjusted fund performance with the help of neural networks and emphasize the influence of macroeconomic conditions on prediction. Compared with well-documented and explored developed mutual fund markets, Chinese mutual funds receive limited attention in the literature. Our study contributes to this strand of literature by answering whether machine learning works in Chinese mutual fund markets.

3. Data

3.1. Mutual funds

Our main data source is Wind, a leading financial data provider in China. We focus on actively managed domestic Chinese equity mutual funds, excluding Qualified Domestic Institutional Investors (QDII) funds, exchange-traded funds (ETFs), and index funds. The sample includes stock, equity oriented, and flexible funds, considering only the A share class for funds with multiple share classes. Additionally, we include only funds with more than 60 % of their portfolios invested in equities. The daily and monthly fund returns are based on the net asset value (NAV) per share, adjusted for splits and dividends. Fig. 1 shows the trend of the fund numbers in our sample. According to our criteria, from the end of January 2010 to the end of March 2024, the number of funds grew from 323 to 3881.

3.2. Mutual fund characteristics

We construct a dataset of 61 fund characteristics based on existing literature, categorized into four types, namely fund fundamentals, performance momentum, holder structure, and trading characteristics.[6] All these fund characteristics are listed in Table 1. For fund fundamentals, we include total net assets (Pástor et al., 2015; Zhu, 2018), fund age (Jones and Mo, 2021), and fund flow volatility (DeMiguel et al., 2023). To capture performance momentum, we follow DeMiguel et al. (2023) and Kaniel et al. (2023) by incorporating various metrics, such as the t-statistics of factor loadings (betas) from rolling-window time-series regressions of fund returns on the six-factor model (Fama-French five-factor model augmented with a momentum factor), information ratio, and Sharpe ratio.[7] Specifically, we apply various time horizons on the regression, including 90, 120, 180, 250, and 750 trading days, aiming to capture momentum effects across different time scales (short-term, medium-term, and long-term). Moreover, given the dominance of retail investors in the Chinese stock market, we include the proportion of institutional investors among fund shareholders (Jiang et al., 2020). Additionally, building on Shang and Sutcliffe (2024), who highlight the influence of fund management company shareholders on mutual fund performance, we also incorporate the proportion of fund company employee ownership in the fund.

Furthermore, Alexander et al. (2007) show that mutual fund performance is linked to trade motivations, including value-, liquidity-, and window-dressing-driven trading. Accordingly, we construct trading-related fund characteristics to reflect these behaviors. First, if a fund manager engages in value-motivated trading, they may target specific individual stocks, leading to a higher dispersion in individual stock transaction volumes. As such, based on disclosed high-volume transaction data for individual stocks in each fund reporting period,[8] we calculate the average and variance of the proportion of buy amounts for each stock relative to the fund's TNA at the beginning of the period, denoted by *TS_Buy_Mean* and *TS_Buy_Std*. In addition, if the fund's asset under management (AUM) increases during a given period because of positive net inflows and the fund executes significant selloffs, it indicates a stronger inclination toward value-driven selling behavior, denoted by *SF*. Besides, to measure window dressing, we use the *Single Backward Rank Gap* (*SBRG*). If a fund holds its disclosed portfolio from the end of the previous reporting period unchanged throughout the next reporting period, this simulated portfolio return is recorded as the *Backward Holding Return*. By ranking the backward holding returns of the target fund against the actual returns of all other funds in descending order, we derive the fund's *Single Backward Simulated Rank* (*SBSR*). For each period, the *SBRG* is calculated as the difference between the actual return rank and the corresponding *SBSR*. Specifically, we calculate the average window-dressing measures across the current and previous reporting periods to mitigate the impact of any bias within a single period, denoted by *SBRG_Mean*. Further, the disclosed income from equity spreads represents the realized gains of a fund, including dividends received and profits from selling stocks.[9] Higher stock spread income indicates that the fund manager has strong stock trading capabilities, effectively reflecting the outcomes of their trading decisions. Thus, we build *TRADE_INCs* by dividing the stock spread income by the fund's TNA at the beginning of the period.

3.3. Summary statistics

Table 2 presents the summary statistics for the fund characteristics listed in Table 1. All continuous variables are truncated at the conventional 1st and 99th percentiles to reduce the potential impact of outliers. Fundamentals such as TNA reveal an average fund size

[6] While our study does not include all fund features identified in prior research, our framework is adaptable to incorporate additional factors.

[7] The returns for size (SMB, small minus big), value (HML, high minus low book-to-market equity), profitability (RMW, robust minus weak), investment (CMA, conservative minus aggressive), and momentum (UMD, up minus down) factors are retrieved from https://www.factorwar.com/data/factor-models/.

[8] Chinese mutual funds need to disclose details of stocks where the fund's cumulative buy and sell amounts exceed 2 % of the fund's initial TNA or fall within the top 20 stocks by transaction value in fund semiannual and annual reports.

[9] Chinese mutual funds need to disclose the income from equity investments - spread income from the purchase and sale of shares in the income statement of fund semiannual and annual reports.

Table 2
Summary Statistics.

Variable	Mean	Std	Min	P25	Median	P75	Max
TNA	1009.342	1543.275	8.134	114.054	365.148	1208.056	10,229.879
Fund_Age	1865.146	1420.311	154.000	729.000	1487.000	2639.000	6402.000
Vol_Flows	12,695.474	22,518.094	0.000	954.936	3960.852	13,246.863	168,686.003
Const_90	−0.002 %	0.074 %	0 %	−0.049 %	−0.004 %	0.043 %	0.226 %
Const_120	−0.002 %	0.066 %	0 %	−0.045 %	−0.005 %	0.038 %	0.202 %
Const_180	−0.001 %	0.057 %	0 %	−0.039 %	−0.003 %	0.035 %	0.174 %
Const_250	0.001 %	0.050 %	0 %	−0.032 %	−0.001 %	0.033 %	0.156 %
Const_750	0.008 %	0.041 %	0 %	−0.019 %	0.009 %	0.035 %	0.121 %
R_Squared_90	0.775	0.150	0.126	0.711	0.811	0.881	0.977
R_Squared_120	0.763	0.155	0.109	0.695	0.800	0.872	0.974
R_Squared_180	0.743	0.160	0.089	0.667	0.782	0.858	0.970
R_Squared_250	0.730	0.162	0.080	0.649	0.769	0.847	0.966
R_Squared_750	0.693	0.162	0.060	0.609	0.726	0.811	0.946
t_Const_90	−0.070	1.087	−2.822	−0.821	−0.069	0.678	2.771
t_Const_120	−0.073	1.103	−2.832	−0.838	−0.090	0.678	2.891
t_Const_180	−0.037	1.130	−2.808	−0.831	−0.066	0.735	2.985
t_Const_250	0.018	1.138	−2.772	−0.778	−0.025	0.782	3.079
t_Const_750	0.326	1.232	−2.576	−0.567	0.291	1.200	3.471
t_MKT_90	13.486	7.174	1.243	8.544	12.050	16.783	44.553
t_MKT_120	15.356	8.067	1.367	9.808	13.811	19.100	50.029
t_MKT_180	18.114	9.468	1.649	11.597	16.411	22.605	58.177
t_MKT_250	20.602	10.821	1.835	13.091	18.785	25.941	65.567
t_MKT_750	28.777	16.243	2.072	16.784	26.323	38.149	90.819
t_SMB_90	−0.022	2.109	−5.558	−1.428	−0.039	1.316	6.758
t_SMB_120	−0.037	2.300	−6.230	−1.559	−0.060	1.417	7.384
t_SMB_180	−0.116	2.585	−7.128	−1.824	−0.147	1.515	8.192
t_SMB_250	−0.195	2.839	−7.761	−2.069	−0.230	1.596	8.871
t_SMB_750	−0.287	3.396	−9.344	−2.503	−0.331	1.825	11.020
t_HML_90	−1.873	1.994	−7.027	−3.216	−1.879	−0.547	4.327
t_HML_120	−2.090	2.185	−7.724	−3.571	−2.076	−0.627	4.774
t_HML_180	−2.381	2.481	−8.652	−4.095	−2.341	−0.676	5.378
t_HML_250	−2.611	2.764	−9.598	−4.542	−2.534	−0.674	5.822
t_HML_750	−3.887	3.989	−14.097	−6.547	−3.460	−0.960	6.718
t_RMW_90	0.040	2.045	−5.435	−1.362	0.051	1.461	5.270
t_RMW_120	0.021	2.301	−6.197	−1.550	0.028	1.615	5.883
t_RMW_180	−0.054	2.739	−7.598	−1.911	−0.040	1.823	6.931
t_RMW_250	−0.177	3.139	−8.601	−2.302	−0.171	1.983	7.851
t_RMW_750	−0.433	4.240	−11.315	−3.129	−0.458	2.176	11.290
t_CMA_90	−0.399	1.698	−4.797	−1.536	−0.380	0.755	4.060
t_CMA_120	−0.482	1.876	−5.480	−1.718	−0.438	0.797	4.308
t_CMA_180	−0.640	2.128	−6.552	−2.017	−0.540	0.841	4.395
t_CMA_250	−0.828	2.364	−7.590	−2.343	−0.709	0.842	4.551
t_CMA_750	−1.128	2.957	−8.926	−3.132	−1.005	0.928	5.843
t_UMD_90	1.301	2.371	−4.567	−0.351	1.222	2.927	7.481
t_UMD_120	1.500	2.578	−4.900	−0.315	1.393	3.269	8.195
t_UMD_180	1.815	2.817	−4.958	−0.239	1.665	3.805	9.154
t_UMD_250	2.096	3.049	−5.012	−0.198	1.887	4.265	10.185
t_UMD_750	3.271	4.315	−6.066	0.079	2.659	6.188	14.871
IR_P6M	0.024	1.441	−3.588	−0.993	0.015	1.024	3.683
IR_P1Y	0.045	1.046	−2.635	−0.674	0.032	0.763	2.725
IR_P3Y	0.109	0.646	−1.627	−0.328	0.119	0.567	1.654
SR_P6M	0.097	1.433	−2.844	−1.021	0.010	1.159	3.758
SR_P1Y	0.166	1.079	−1.972	−0.680	0.050	0.969	2.854
SR_P3Y	0.437	0.517	−0.784	0.063	0.461	0.813	1.652
Institute	20.716	27.326	0.000	0.380	7.140	32.580	99.840
Worker	0.240	0.593	0.000	0.002	0.040	0.180	5.770
TS_Buy_Mean	3.639	2.418	0.436	2.355	3.247	4.148	23.452
TS_Buy_Std	1.931	2.179	0.174	0.845	1.327	2.135	22.553
SF	1.826	1.741	0.059	0.728	1.302	2.285	13.456
SBRG_Mean	125.289	470.209	−1175.000	−138.000	73.500	357.000	1758.000
TRADE_INCs	0.036	0.123	−0.247	−0.046	0.022	0.109	0.548

This table reports monthly summary statistics, including mean, standard deviation, minimum, 25th, 50th, and 75th percentiles, and maximum for the Chinese mutual fund characteristics in Table 1.

of 1009.34 million RMB, with significant variability, while fund age averages 1865 days, ranging from 154 to over 6400 days, reflecting both young and mature funds. In addition, Sharpe and Information Ratios improve over longer periods, with *SR_P3Y* and *IR_P3Y* averaging 0.44 and 0.11, respectively. The holder structure reveals an average institutional shareholding of 20.72 %, while

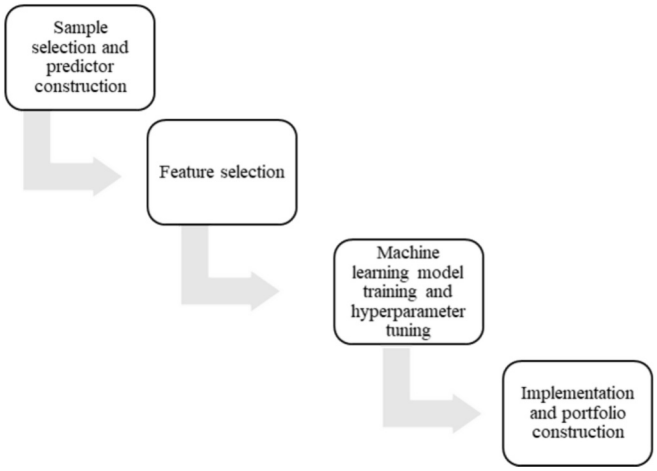


Fig. 2. Overview of Methodology.
Fig. 2 shows the structured multi-step methodology used in this study.

employee shareholding is minimal, with a mean of 0.24 %. Overall, the table illustrates the heterogeneity and dynamic nature of fund attributes in the Chinese market.

4. Methodology

Our methodology follows a structured multi-step process. Fig. 2 shows a graphical representation. We begin with sample selection and predictor construction, as discussed in Section 3, where relevant datasets are employed, and predictive variables (fund characteristics) are constructed based on theoretical and empirical considerations.[10] This is followed by feature selection, which refines the set of predictors by employing statistical techniques or machine learning-based importance measures to isolate the most relevant features, thus reducing model complexity and increasing interpretability. Subsequently, we proceed to machine learning model training and hyperparameter tuning, wherein machine learning methods are implemented. Their hyperparameters are tuned through cross-validation. Finally, in the implementation and fund portfolio construction phase, the trained models are applied to generate predictions resulting in winning funds, which are then used to construct fund portfolios. In this section, we first introduce the employed machine learning methods and then describe the specific implementation process for these methods.

4.1. Machine learning methods

The unique characteristics of the Chinese mutual fund market, including high retail investor participation, distinctive regulatory features from developed markets, and significant volatility (Cornell et al., 2020; Zhang et al., 2024), present challenges for fund performance prediction. Yet, these complexities make machine learning approaches suitable for this task. Given the advantages of tree-based machine learning models, including their strong performance and explainability demonstrated in prior studies (Gu et al., 2020; DeMiguel et al., 2023), we utilize three tree-based ensemble machine learning models in this study, namely Random Forest, eXtreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM), which are described in the Appendix. These models offer a strong balance of interpretability, efficiency, and performance, making them well-suited for financial applications where transparency and robustness are valued. Since the core objective of this study is to assess the applicability and predictive power of machine learning models in fund selection in China, rather than to optimize performance through highly complex architectures such as deep learning, we do not involve other machine learning approaches. Nevertheless, we acknowledge the growing relevance of deep learning techniques in this space and consider this a promising direction for future research, which is discussed in Section 6.

4.2. Machine learning implementation

4.2.1. Model training and out-of-sample test

Following DeMiguel et al. (2023), we use a K-fold cross-validation approach for model training. In K-fold cross-validation, the data is divided into K subsets for each training iteration. Of these, K-1 subsets are used as the training set, while the remaining subset is used as the validation set. Each subset takes turns as the validation set. The prediction error (or cross-validation error) is then calculated for each set of hyperparameter values of the machine learning methods on the validation set. After repeating this process for all K folds, the average cross-validation error is computed across the folds, and the hyperparameter configuration that minimizes this average error is

[10] We normalize these variables to have a zero mean and unit standard deviation when employing machine learning methods.

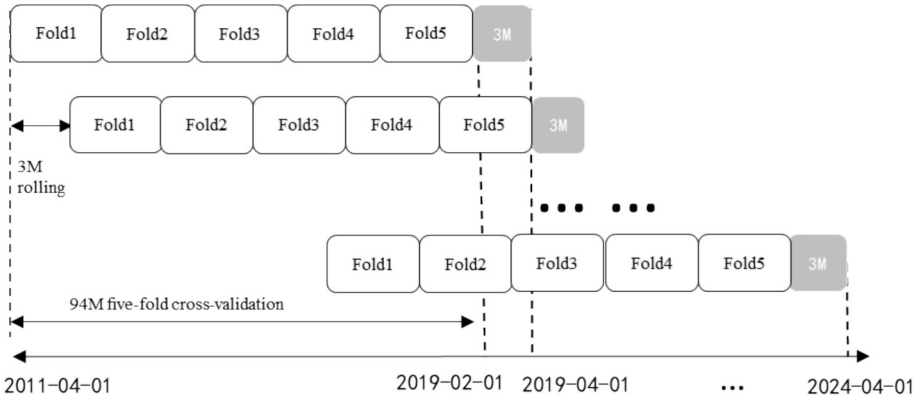


Fig. 3. Machine Learning Implementation.

Fig. 3 shows the training process of machine learning methods.

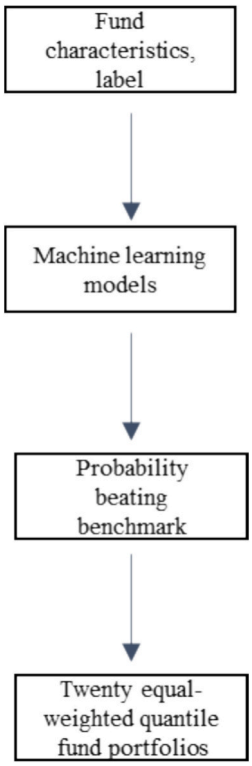


Fig. 4. Fund Portfolio Construction.

Fig. 4 shows how to construct the fund portfolio based on three machine learning methods.

selected.

Fig. 3 illustrates the process.[11] Our training starts in April 2011, with a fixed in-sample training and validation period of 94 months combined and an out-of-sample testing period of 3 months (from February 2019 to March 2019, represented in dark grey). After each round of training, the window rolls forward for 3 months to continue the process. Consequently, our out-of-sample testing period spans from February 2019 to March 31, 2024.

[11] To clarify, hyperparameter tuning is conducted within each training window using 5-fold cross-validation. Specifically, for every 94-month in-sample period, we apply 5-fold cross-validation solely within that window to determine the optimal hyperparameters. This tuning process is repeated independently as the window rolls forward every 3 months.

4.2.2. Machine learning fund portfolio construction

We formulate the fund-selection task as a classification problem rather than a regression problem, as the literature suggests classification models can outperform regression models in predicting financial market behavior (Krauss et al., 2017; Zhao et al., 2025). More importantly, the classification-based framework employed in our study offers the advantage of providing the probability that a fund will outperform the benchmark, allowing fund managers to make more informed and confident decisions. By estimating the likelihood of outperformance, rather than predicting the exact magnitude of excess returns, the classification approach aligns with the decision-making process of fund managers of FoFs, who are typically focused on identifying funds that are more likely to outperform their peer benchmarks (Sensoy, 2009). This probabilistic output enables a more strategic selection of funds, with an enhanced ability to assess risk and potential for success.[12] Consequently, rather than focusing on risk-adjusted realized alphas or abnormal returns (e.g., DeMiguel et al., 2023; Kaniel et al., 2023), our study leverages machine learning methods to select funds with a high likelihood of exceeding the benchmark. Specifically, for the benchmark, we use the Wind Equity Oriented Fund Index (885,001.WI). This index includes funds that have been established for over three months and whose equity asset allocation either has an upper limit of no less than 75 % or a lower limit of no less than 50 %. It can reflect the overall performance of equity funds. Additionally, as shown in Fig. A2 of the Online Appendix, this index has consistently surpassed key market indices, demonstrating strong performance over time. Hence, it is a rigorous benchmark to evaluate the effectiveness of the machine learning-based fund portfolio.

Moreover, we construct binary labels for this classification problem based on the relative performance of each fund compared to the benchmark. If a fund's monthly return exceeds the benchmark's monthly return, the label is assigned a value of 1; otherwise, it is assigned a value of 0. As shown in Table A1 of the Online Appendix, the binary labels are well balanced throughout the sample period. The average monthly ratios of each class (0 and 1) fluctuate closely around 50 %, with no persistent imbalance. Given this observed balance, class imbalance is not an issue in our dataset.

Further, after training the models on the labels and fund characteristics, we apply the trained models to the test sample to predict the probability of each fund outperforming the benchmark in the following month. Based on these predicted probabilities, we segment the funds into 20 equal-weighted quantile portfolios. Portfolio 0 (Top) consists of funds in the top 5 % of predicted probabilities, representing the funds most likely to outperform, while Portfolio 19 (Bottom) contains funds in the lowest 5 %. Consistent with our expectations, Portfolio 0 is anticipated to deliver the strongest performance. Fig. 4 illustrates the process of machine learning-based fund portfolio construction.

5. Empirical results

5.1. Feature selection

We apply feature selection to filter fund characteristics, which can reduce model complexity, avoid dimensional disasters caused by too many features, and increase model interpretability. Using data from March 2011 to December 2018, the selection process follows several rules to ensure robustness and relevance.[13] First, at least one variable is included in each category to ensure feature coverage. Second, predictors are evaluated through univariate sorts: each month, mutual funds are sorted by a specific characteristic, and 20 equal-weighted quantile portfolios are formed. The long (short) quantile portfolio consists of funds with the highest (lowest) values of the characteristic. Excess returns from the long-short portfolio are then calculated to assess the variable's ability to differentiate between long and short positions. Third, the information coefficient (IC), defined as Spearman's rank correlation coefficient between fund returns and predictor variables, a widely used measure in practice, must have an absolute value of at least 0.03 to demonstrate predictive power. The results of long-short portfolios and IC tests are reported in Table A2 of the Online Appendix. Lastly, variables are prioritized based on their feature importance (presented in Table A3) rankings within machine learning models during the evaluation period. Following this process, we identify and select 22 fund characteristics, as detailed in Table 3.

According to Table 3, the most frequently selected predictors are the *Const* and *t_Const* variables, which represent the alpha (fund's historical risk-adjusted performance) and its t-statistic (statistical reliability) from the regressions of fund returns on the six-factor model (Fama-French five-factor plus momentum) over different horizons (90, 120, 180, 250, and 750 trading days). Their consistent selection highlights the predictive importance of performance momentum across both short- and long-term windows, in line with prior studies, such as DeMiguel et al. (2023) and Kaniel et al. (2023). Moreover, out of a much larger set of candidate features, only 22 fund characteristics are retained through embedded feature selection by the tree-based models (Random Forest, XGBoost, LightGBM), which prioritize variables with the greatest contribution to reducing out-of-sample prediction error. The dominance of *Const* and *t_Const* terms suggests that historical alpha signals, especially those with statistical strength, remain powerful and stable predictors of

[12] Nevertheless, a comparison with regression-based models can shed light on the relative strengths and weaknesses of the classification-based framework adopted in our study. Thus, we also provide a performance summary of regression-based models in Table A4 of the Online Appendix, and the results discussion is presented in Section 5.2.

[13] We acknowledge that feature importance may vary over time, and our static feature ranking may underrepresent time-sensitive predictors. In our current setup, feature selection is conducted once using data from March 2011 to December 2018, rather than being reapplied in each rolling window. This choice is mainly driven by considerations of computational efficiency. Additionally, the selection window covers more than 50 % of the full sample period (until March 2024), which, to some extent, helps ensure that the chosen features broadly reflect the underlying data characteristics. However, this approach may overlook evolving patterns in feature relevance. Future extensions could incorporate dynamic feature selection strategies to better adapt to evolving market conditions and capture temporal changes in predictor importance.

Table 3
Feature Selection.

Type	Variable
Fundamentals	TNA
	Vol_Flows
Performance Momentum	t_Const_90
	t_Const_120
	t_Const_180
	t_Const_250
	t_Const_750
	Const_90
	Const_120
	Const_180
	Const_250
	Const_750
	IR_P1Y
	IR_P3Y
	SR_P1Y
	SR_P3Y
Holder Structure	Institute
	Worker
Trading Characteristics	TS_Buy_Mean
	SF
	SBRG_Mean
	TRADE_INCs

This table lists the 22 fund characteristics after feature selection.

future fund performance. This selection reflects the underlying economic intuition that persistent outperformance and the confidence in that outperformance are critical indicators for fund selection.

5.2. Performance of machine learning portfolios

Following the machine learning portfolio construction strategy in Fig. 4, we utilize the first 94 months of data on one-month-ahead fund labels (binary indicators denoting whether a fund outperforms the benchmark fund index from April 2011 to January 2019) along with one-month-lagged selected fund characteristics (from March 2011 to December 2018) to train three tree-based machine learning models. Then, using fund characteristics data from January to March 2019, which are not employed in the training process, we predict whether each fund outperforms the benchmark index in February, March, and April 2019, respectively. For each of these three out-of-sample months, we create 20 equally weighted portfolios based on predicted probabilities, with the first portfolio comprising funds in the top 5 % of the predicted probability distribution. The returns of these portfolios, net of expenses, fees, loads, and transaction costs, are tracked for the respective month.[14] If a fund exits from the sample during this period, its allocation is equally redistributed among the remaining funds in the portfolio. Every three months, we roll the training sample forward by three months, retrain the models on the updated data, and make new predictions for the subsequent three months. New fund portfolios are then constructed, and their net returns are tracked. This iterative process generates a time series of monthly out-of-sample net returns for the 20 fund portfolios, covering February 2019 through March 2024 (a total of 62 months).

To assess the out-of-sample performance of the 20 fund portfolios, we conduct a time-series regression of the 62 out-of-sample monthly portfolio excess returns against the returns of risk factors. Following Li and Rao (2023), we adopt the CH-3 and CH-4 models of Liu et al. (2019), as these models are better suited for explaining anomalies in the Chinese stock market compared to the traditional Fama-French three-factor and Carhart four-factor models. The specifications using the CH-4 model are given as follows:

$$r_t - r_{ft} = \alpha_t + \beta_1 MKT_t + \beta_2 SMB_t + \beta_3 VMG_t + \beta_4 PMO_t + e_t, \# \quad (1)$$

where r_t is the net fund portfolio return for the month t, r_{ft} is the risk-free rate (the one-year deposit rate divided by 12), MKT_t, SMB_t, VMG_t, and PMO_t represents four factors for market, size, value, and turnover, and α_t represent the risk-adjusted fund portfolio performance or realized alpha.[15] Statistical inference is conducted using Newey and West (1987) standard errors, which correct for autocorrelation and heteroskedasticity in the residuals.

Table 4 reports the performance of the 20 portfolios constructed using three machine learning methods: Random Forest, XGBoost,

[14] The net fund returns used in our analysis already account for expense-related fees, as they are reported net of management and operational fees. Regarding transaction costs, we conservatively apply a rate of 0.75 % of net assets. This level reflects typical market conditions in China, where the average redemption fee is around 0.5 % for holding periods of 30 days or more, and sales loads are often discounted to 0.15 % or less for large purchase amounts (e.g., over 1 million yuan).

[15] The risk-free rate and the factor returns are obtained from the website of Robert F. Stambaugh (https://finance.wharton.upenn.edu/~stambaug/).

and LightGBM. Portfolios are ranked by their predicted probabilities of outperforming the benchmark fund index, with Portfolio 0 representing the top 5 % of funds and Portfolio 19 representing the bottom 5 %. Panel A summarizes the performance of portfolios constructed using the Random Forest model. High-ranking portfolios, such as Portfolio 0, achieve superior performance with an annualized return of 15.43 %, a Sharpe ratio of 0.65, and a CH-4 alpha of 0.017 (significant at the 1 % level). In contrast, Portfolio 19 shows weaker performance, with an annualized return of 8.38 %, a Sharpe ratio of 0.42, and a CH-4 alpha of 0.007 (significant at the 10 % level). The long-short strategy (Portfolio 0 minus Portfolio 19) delivers an annualized return of 7.02 %. Panel B (XGBoost) and Panel C (LightGBM) present similar patterns: top-ranked portfolios substantially outperform bottom-ranked ones, and the strategies generate annualized returns above 7 %.

The results of Table 4 demonstrate the effectiveness of machine learning methods in predicting fund performance and constructing outperforming portfolios.[16] Among the three methods, LightGBM consistently achieves the best results. In addition, compared with the results in Table A4 of the Online Appendix, we find the classification-based approach consistently delivers superior performance over the regression-based framework. For example, the long-short annualized returns from the classification models range from 7.02 % to 8.07 %, compared to 3.17 % to 6.00 % for the regression models. Similarly, Sharpe ratios for classification models are higher (up to 0.87), and they exhibit lower maximum drawdowns. These findings suggest that our classification framework not only enables fund managers to make probabilistic decisions but also provides a more effective framework for fund selection and portfolio construction.

Moreover, Fig. 5 shows the annualized excess returns of portfolios constructed using three machine learning models relative to the benchmark fund index. Portfolio 0, representing the top 5 % of funds with the highest predicted probabilities of outperforming the benchmark, consistently achieves the highest excess returns across all models, with LightGBM exhibiting the strongest performance at nearly 6 %. Portfolios ranked lower (1–9) show diminishing but generally positive returns, while the lowest-ranked portfolios (10–19) underperform significantly, with Portfolio 19 (Bottom 5 %) showing the most negative excess returns across all methods. These results again highlight the machine learning models' ability to rank funds by their potential to outperform the benchmark accurately.

Since Portfolio 0 includes funds with the highest predicted probabilities of outperforming the benchmark fund index, it represents the investable portfolio for investors. Fig. 6 presents its cumulative performance, alongside the benchmark fund index, over the out-of-sample investment horizon from February 2019 to March 2024. Assuming an initial wealth of 1, all three machine learning-based portfolios demonstrate significant growth, consistently outperforming the benchmark index throughout the period. Likewise, among the models, LightGBM achieves the highest cumulative return, followed closely by Random Forest and XGBoost, reflecting the effectiveness of these algorithms in identifying high-performing funds. In contrast, the benchmark index exhibits comparatively modest growth, underscoring the value of machine learning-driven portfolio construction in achieving superior investment outcomes. However, we can see a drawdown of the portfolios post-COVID due to the decline of the entire Chinese fund market (Zhou and Liu, 2024).

5.3. *Machine learning combination*

Combining forecasts from multiple prediction methods often outperforms relying on a single "best forecast." This approach integrates diverse drivers of the data-generating process and helps reduce uncertainties related to model structure and parameter estimation (Wang et al., 2023). Previous research has demonstrated that forecast combination methods are particularly effective in improving return predictability (Rapach et al., 2010; Bali et al., 2020). As such, we also employ forecast combinations in our study. As shown in Fig. 7, each model generates independent probability estimates of fund performance surpassing the benchmark fund index, which are then averaged to produce a final probability score, leveraging the diverse strengths of each algorithm while mitigating individual model biases.

Fig. 8 shows the annualized excess returns of 20 quantile portfolios constructed using the machine learning forecast combination relative to the benchmark fund index. While both Fig. 5 and Fig. 6 highlight the superior performance of LightGBM compared to Random Forest and XGBoost, the quantile portfolios derived from the forecast combination demonstrate improved monotonicity compared to those constructed using individual machine learning methods, particularly evident in Portfolios 0 to 4 and 17 to 19. This demonstrates the advantages of the forecast combination approach.

Table 5 reports the performance of the 20 portfolios constructed using the machine learning combination framework. The top-ranked portfolio, Portfolio 0, demonstrates the strongest performance across all metrics, achieving a total return of 114.40 %, an annualized return of 15.91 %, and a Sharpe ratio of 0.68. Its annualized return over the benchmark index is 4.82 %, and its CH-4 alpha is 0.017, significant at the 1 % level. This highlights the ability of the combined machine learning approach to effectively identify funds with superior performance potential. Lower-ranked portfolios, such as Portfolio 19, exhibit significantly poorer performance, with a total return of 48.96 %, an annualized return of 8.02 %, and a negative annualized return over the index of −3.08 %. Moreover, the long-short strategy, which involves investing in Portfolio 0 and shorting Portfolio 19, achieves an annualized return of 7.66 %.

While the machine learning combination that takes the simple averaging of predictions from Random Forest, XGBoost, and LightGBM provides stable performance and improved monotonicity for quantile fund portfolios, it does not consistently outperform

[16] We have also investigated the holding similarity of funds in Portfolio 0 following Elton et al. (2007). Our analysis shows that while there is some level of commonality, the degree of similarity does not suggest excessive concentration or redundancy in fund selection. For example, in April 2023, based on the stock holdings of the top 2 funds in Portfolio 0 at the end of 2022, the calculated holding similarity of the two funds is only 0.55 %. This indicates that our framework is capable of identifying high-performing funds with sufficiently diverse exposures, thereby maintaining the benefits of diversification. The detailed results are upon request.

Table 4
Performance of Machine Learning Portfolios.

Panel A: Quantile Fund Portfolio Performance - Random Forest						
Portfolio	Total Return	Annualized Return	Sharpe Ratio	Annualized Return Over the Index	CH3 Alpha	CH4 Alpha
0	109.90 %	15.43 %	0.65	4.48 %	0.016*** (3.85)	0.017*** (3.65)
1	78.83 %	11.91 %	0.50	1.32 %	0.013** (3.11)	0.014** (3.0)
2	91.10 %	13.35 %	0.57	2.56 %	0.014** (3.22)	0.015** (3.09)
3	78.57 %	11.87 %	0.51	1.17 %	0.013** (2.83)	0.013** (2.77)
4	82.02 %	12.29 %	0.54	1.43 %	0.013** (3.10)	0.013** (2.97)
5	68.24 %	10.59 %	0.48	−0.23 %	0.011** (2.87)	0.012** (2.70)
6	73.17 %	11.21 %	0.51	0.37 %	0.011** (2.86)	0.012** (2.73)
7	55.09 %	8.86 %	0.39	−1.70 %	0.010* (2.51)	0.010* (2.44)
8	71.94 %	11.06 %	0.51	0.17 %	0.011** (2.88)	0.012** (2.75)
9	70.63 %	10.89 %	0.50	0.03 %	0.011** (2.87)	0.012** (2.74)
10	64.64 %	10.13 %	0.47	−0.77 %	0.010* (2.58)	0.011* (2.53)
11	71.85 %	11.05 %	0.53	−0.01 %	0.011** (2.70)	0.011* (2.65)
12	59.48 %	9.45 %	0.45	−1.43 %	0.010* (2.61)	0.010* (2.57)
13	65.60 %	10.25 %	0.49	−0.72 %	0.010* (2.64)	0.011* (2.61)
14	69.23 %	10.72 %	0.52	−0.41 %	0.010** (2.78)	0.011** (2.67)
15	60.64 %	9.61 %	0.46	−1.32 %	0.009* (2.35)	0.009* (2.22)
16	60.92 %	9.64 %	0.47	−1.41 %	0.009* (2.21)	0.009* (2.11)
17	71.69 %	11.03 %	0.55	−0.25 %	0.010** (2.72)	0.010** (2.67)
18	61.10 %	9.67 %	0.46	−1.33 %	0.009* (2.38)	0.009* (2.24)
19	51.54 %	8.38 %	0.42	−2.76 %	0.007* (2.20)	0.007* (2.09)
Long-Short	41.95 %	7.02 %	0.79	−7.06 %	0.009*** (3.35)	0.010*** (3.60)
Panel B: Quantile Fund Portfolio Performance - XGBoost						
0	114.08 %	15.87 %	0.68	4.77 %	0.016*** (3.73)	0.017*** (3.59)
1	84.69 %	12.61 %	0.54	1.85 %	0.014** (3.23)	0.015** (3.13)
2	74.33 %	11.36 %	0.48	0.77 %	0.013** (2.81)	0.013** (2.70)
3	82.50 %	12.35 %	0.53	1.59 %	0.013** (3.08)	0.014** (2.97)
4	77.19 %	11.71 %	0.52	0.87 %	0.013** (2.86)	0.013** (2.78)
5	77.53 %	11.75 %	0.53	0.87 %	0.013** (3.10)	0.013** (2.95)
6	61.05 %	9.66 %	0.43	−0.98 %	0.010** (2.67)	0.011* (2.58)
7	74.65 %	11.39 %	0.52	0.52 %	0.012** (3.08)	0.012** (2.95)
8	75.24 %	11.47 %	0.52	0.55 %	0.012** (3.06)	0.012** (2.93)
9	60.01 %	9.52 %	0.43	−1.19 %	0.010* (2.43)	0.010* (2.30)
10	69.06 %	10.70 %	0.50	−0.29 %	0.011**	0.011**

(*continued on next page*)

Table 4 (*continued*)

					(2.78)	(2.68)
11	55.31 %	8.89 %	0.41	−1.87 %	0.009*	0.010*
					(2.37)	(2.30)
12	61.51 %	9.72 %	0.46	−1.22 %	0.009*	0.010*
					(2.44)	(2.47)
13	77.88 %	11.79 %	0.57	0.62 %	0.011**	0.012**
					(2.97)	(2.92)
14	64.18 %	10.07 %	0.47	−0.85 %	0.010*	0.010*
					(2.52)	(2.37)
15	64.60 %	10.13 %	0.48	−0.87 %	0.010*	0.010*
					(2.56)	(2.41)
16	76.92 %	11.67 %	0.57	0.44 %	0.011*	0.011*
					(2.65)	(2.56)
17	68.20 %	10.59 %	0.51	−0.54 %	0.010*	0.010*
					(2.59)	(2.51)
18	53.57 %	8.66 %	0.43	−2.38 %	0.008*	0.008*
					(2.22)	(2.11)
19	45.33 %	7.50 %	0.38	−3.52 %	0.006	0.006
					(1.93)	(1.83)
Long-Short	49.35 %	8.07 %	0.87	−6.26 %	0.010***	0.011***
					(3.24)	(3.62)
Panel C: Quantile Fund Portfolio Performance - LightGBM						
					0.017***	0.018***
0	119.58 %	16.44 %	0.71	5.31 %	(3.91)	(3.77)
					0.013**	0.014**
1	79.37 %	11.97 %	0.50	1.34 %	(3.02)	(2.91)
					0.014**	0.014**
2	86.53 %	12.82 %	0.55	2.03 %	(3.17)	(3.07)
					0.013**	0.013*
3	75.50 %	11.50 %	0.50	0.82 %	(2.73)	(2.66)
					0.012**	0.012**
4	68.94 %	10.68 %	0.48	−0.10 %	(2.88)	(2.75)
					0.012**	0.013**
5	75.99 %	11.56 %	0.52	0.67 %	(3.04)	(2.91)
					0.013**	0.013**
6	83.69 %	12.49 %	0.55	1.66 %	(3.23)	(3.10)
					0.010**	0.011*
7	64.04 %	10.05 %	0.45	−0.69 %	(2.76)	(2.65)
					0.010*	0.011*
8	62.42 %	9.84 %	0.45	−0.91 %	(2.51)	(2.40)
					0.011**	0.012**
9	70.39 %	10.86 %	0.50	−0.04 %	(2.98)	(2.80)
					0.011**	0.011**
10	70.00 %	10.82 %	0.51	−0.15 %	(2.74)	(2.67)
					0.009*	0.010*
11	57.95 %	9.25 %	0.43	−1.58 %	(2.53)	(2.47)
					0.010*	0.011*
12	63.08 %	9.93 %	0.47	−1.04 %	(2.60)	(2.54)
					0.011**	0.011*
13	71.80 %	11.04 %	0.53	−0.05 %	(2.66)	(2.57)
					0.010**	0.010*
14	64.49 %	10.11 %	0.48	−0.84 %	(2.73)	(2.62)
					0.010*	0.010*
15	70.47 %	10.87 %	0.52	−0.20 %	(2.55)	(2.43)
					0.009*	0.010*
16	66.12 %	10.32 %	0.49	−0.70 %	(2.48)	(2.39)
					0.009*	0.009*
17	58.79 %	9.36 %	0.46	−1.69 %	(2.23)	(2.14)
					0.008*	0.008*
18	56.47 %	9.05 %	0.45	−2.01 %	(2.40)	(2.27)
					0.007*	0.007*
19	52.27 %	8.48 %	0.42	−2.61 %	(2.13)	(2.04)
					0.010***	0.010***
Long-Short	46.57 %	7.68 %	0.86	−6.60 %	(3.45)	(3.81)

This table reports the performance metrics for the 20 fund portfolios constructed using three machine-learning methods: Random Forest (Panel A), XGBoost (Panel B), and LightGBM (Panel C). The metrics include total return, annualized return, Sharpe ratio, annualized return over the fund market index, and alphas relative to the CH3 and CH4 factor models, with corresponding t-statistics in parentheses. Portfolio returns are net of all costs. Statistical significance is denoted by one, two, and three asterisks at the 10 %, 5 %, and 1 % levels, respectively. The table also includes the long-short (0–19) portfolio, representing the performance difference between the highest and lowest quantile portfolios.

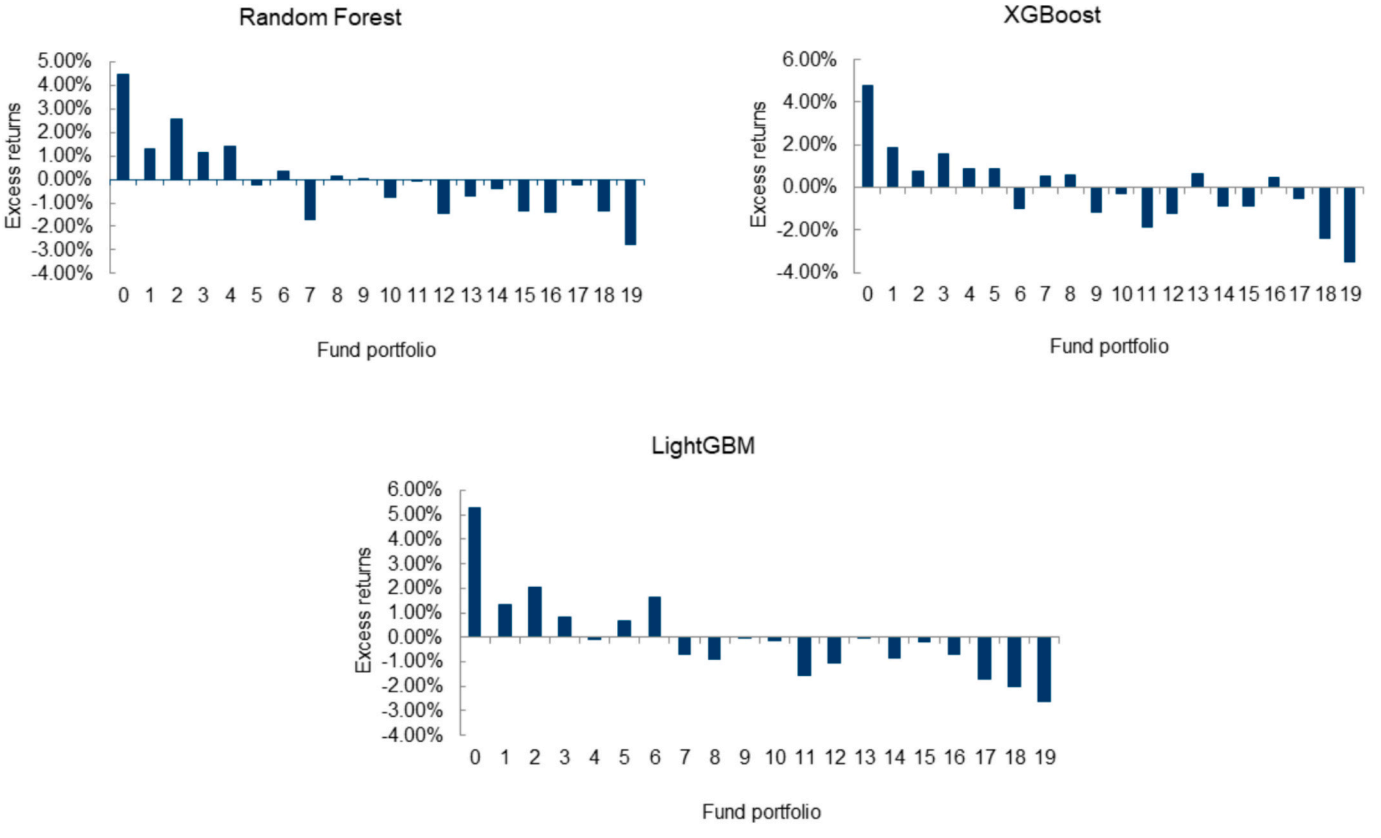


Fig. 5. Annualized Excess Returns of Machine Learning Fund Portfolios.
Fig. 5 shows the annualized excess returns of fund portfolios constructed using Random Forest, XGBoost, and LightGBM over the Wind Equity Oriented Fund Index. From left to right on the x-axis, each chart displays 20 fund portfolios, labeled from Portfolio 0 to Portfolio 19. Portfolio 0 (*Top*) includes funds in the top 5 % of predicted probabilities, those most likely to outperform the benchmark, whereas Portfolio 19 (*Bottom*) consists of funds in the lowest 5 %, representing those least likely to outperform.

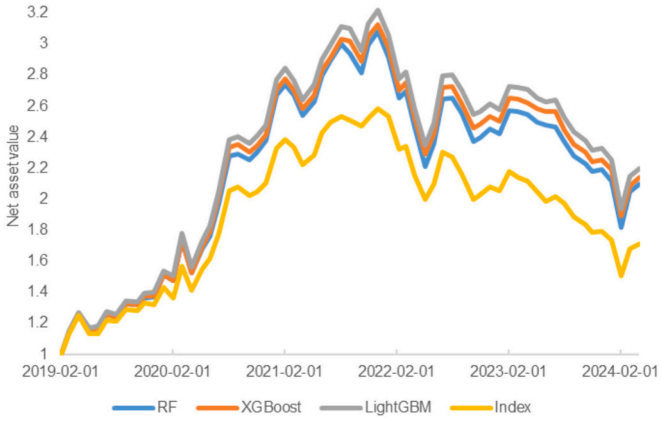


Fig. 6. Performance over the Investment Horizon.
Fig. 6 shows the performance (net asset value assuming an initial wealth of 1) of portfolios constructed using Random Forest, XGBoost, LightGBM, and the benchmark fund index over the investment horizon.

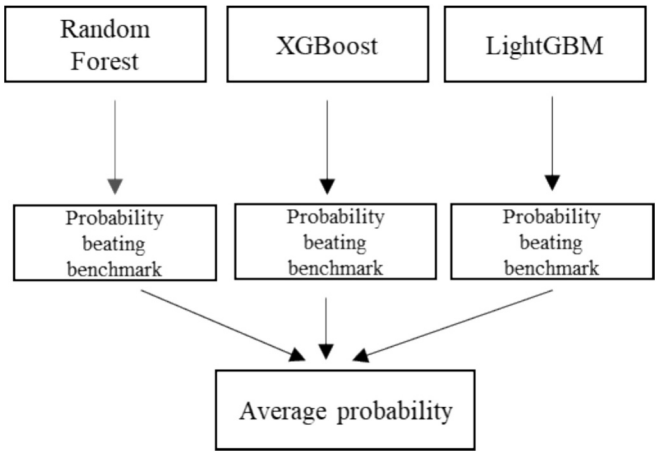


Fig. 7. Machine Learning Combination.

the strongest individual model across all evaluation metrics. We also calculate the maximum drawdown of portfolios from this method and find that, although it is slightly smaller than that of some portfolios constructed using each of the three individual models, the improvement in downside risk is marginal.[17] One potential reason for this is the lack of diversity among the base models (Kang et al., 2022), as all three algorithms used are tree-based ensemble methods. These models have been selected due to their strong predictive performance and interpretability, but their similarity may limit the benefits of model averaging. However, the forecast combination framework employed here significantly enhances the scalability of our machine learning-based fund selection strategy. This framework allows for the integration of additional machine learning techniques, making it adaptable to various predictive models and ensuring more robust and consistent portfolio construction. The combination of multiple models can reduce the risk of relying on any single model's biases, thus improving the overall forecasting accuracy. Moreover, in terms of the forecast combination process, a more sophisticated ensemble approach, such as weighted averaging and stacking, can be employed to enhance the effectiveness of the combination compared to the simple averaging method (Wang et al., 2023).

6. Conclusions

This study demonstrates the potential of machine learning in addressing the complexities of mutual fund selection in emerging markets, with a specific focus on the Chinese mutual fund market. By leveraging advanced machine learning techniques such as Random Forest, XGBoost, and LightGBM alongside an integrated combination framework, we establish a robust methodology for

[17] For example, the maximum drawdown of the long-short portfolio (Portfolio 0 - Portfolio 19) for Random Forest, XGBoost, LightGBM, and machine learning combination is 11.44 %, 9.67 %, 10.94 %, and 10.33 %, respectively.

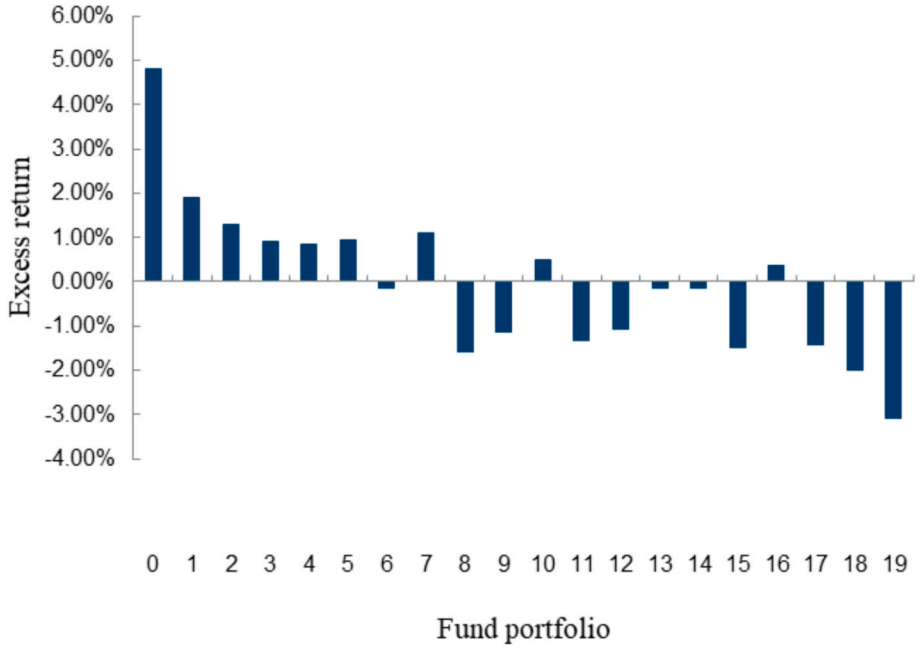


Fig. 8. Annualized Excess Returns of Fund Portfolios by Machine Learning Combination.
Fig. 8 shows the annualized excess returns of fund portfolios constructed using the machine learning forecast combination over the Wind Equity Oriented Fund Index. From left to right on the x-axis, each chart displays 20 fund portfolios, labeled from Portfolio 0 to Portfolio 19. Portfolio 0 (*Top*) includes funds in the top 5 % of predicted probabilities, those most likely to outperform the benchmark, whereas Portfolio 19 (*Bottom*) consists of funds in the lowest 5 %, representing those least likely to outperform.

identifying outperforming funds. The results affirm the predictive strength of these models, offering valuable insights for fund managers, domestic investors, and foreign investors seeking exposure to Chinese assets. The research contributes to the expanding body of literature on fund performance prediction by applying machine learning in an underexplored market, shedding light on the unique characteristics of the Chinese mutual fund industry, including retail investor dominance and market inefficiencies. Practically, this study can provide FoF fund managers with systematic, data-driven tools to optimize portfolio construction, while offering fund investors a reliable framework for identifying high-performing Chinese mutual funds. The proposed combination framework can bring robustness by integrating predictions from multiple models, thereby reducing individual model biases. Additionally, policymakers can also draw on these findings to foster technological integration in mutual fund markets, promoting efficiency and innovation.

While our study highlights the advantages of machine learning in mutual fund selection, it is not without limitations. A key constraint is the scope of features included in the analysis, as we rely primarily on fund-level characteristics and exclude other predictors identified in more recent studies and broader macroeconomic variables. Incorporating such variables in future research could provide a more comprehensive understanding of how external factors influence fund performance and machine learning predictions. Additionally, the study does not differentiate fund performance across varying market conditions. Investigating the robustness and adaptability of machine learning techniques during different market phases, such as bull, bear, or volatile markets, would offer deeper insights into their predictive capabilities under diverse economic scenarios. Moreover, while this research prioritizes tree-based models for their relative interpretability, the potential of more advanced techniques, such as predictors derived from textual analysis of fund disclosures and manager commentary, deep learning, and large language models (LLMs), remains underexplored. As advancements in explainability for deep learning evolve, future studies could evaluate whether these models provide additional predictive power without sacrificing transparency. This exploration would help strike a balance between model accuracy and usability in financial decision-making. In addition, in this study, we adopt a simple averaging method to combine predictions from the three tree-based models, and this approach represents only a basic form of model aggregation. The combination process could be enhanced in future research through the use of more sophisticated ensemble techniques, such as weighted averaging based on model performance and stacking. These methods allow for more adaptive and efficient integration of model outputs, potentially improving overall predictive accuracy.

CRediT authorship contribution statement

Haoran Wu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zhiwei Gao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Boyang Nie:**

Table 5
Performance of Portfolios Using Machine Learning Combination.

Portfolio	Total Return	Annualized Return	Sharpe Ratio	Annualized Return Over the Index	CH3 Alpha	CH4 Alpha
					0.017***	0.017***
0	114.40 %	15.91 %	0.68	4.82 %	(3.78)	(3.68)
					0.014**	0.015**
1	84.65 %	12.60 %	0.53	1.91 %	(3.18)	(3.10)
					0.013**	0.014**
2	78.69 %	11.89 %	0.50	1.28 %	(2.76)	(2.72)
					0.013**	0.013**
3	76.64 %	11.64 %	0.51	0.92 %	(2.86)	(2.80)
					0.012**	0.013**
4	77.12 %	11.70 %	0.52	0.85 %	(2.87)	(2.78)
					0.013**	0.013**
5	78.22 %	11.83 %	0.53	0.94 %	(3.08)	(2.98)
					0.011**	0.012**
6	68.66 %	10.65 %	0.48	−0.14 %	(2.94)	(2.85)
					0.012**	0.012**
7	79.32 %	11.97 %	0.54	1.10 %	(3.06)	(2.93)
					0.010*	0.010*
8	56.81 %	9.10 %	0.41	−1.58 %	(2.54)	(2.48)
					0.010*	0.011*
9	61.04 %	9.66 %	0.44	−1.13 %	(2.52)	(2.44)
					0.012**	0.012**
10	75.26 %	11.47 %	0.53	0.51 %	(2.81)	(2.75)
					0.009*	0.010*
11	60.71 %	9.62 %	0.46	−1.33 %	(2.43)	(2.42)
					0.010*	0.011*
12	62.77 %	9.89 %	0.47	−1.07 %	(2.55)	(2.60)
					0.011**	0.011**
13	70.98 %	10.94 %	0.52	−0.13 %	(2.81)	(2.73)
					0.011**	0.011*
14	70.44 %	10.87 %	0.51	−0.15 %	(2.71)	(2.62)
					0.009*	0.009*
15	59.74 %	9.49 %	0.45	−1.48 %	(2.33)	(2.23)
					0.010*	0.011*
16	75.68 %	11.52 %	0.55	0.37 %	(2.63)	(2.55)
					0.009*	0.009*
17	61.29 %	9.69 %	0.48	−1.44 %	(2.30)	(2.25)
					0.008*	0.008*
18	55.85 %	8.97 %	0.43	−2.01 %	(2.27)	(2.17)
					0.007*	0.007
19	48.96 %	8.02 %	0.40	−3.08 %	(2.07)	(1.98)
					0.010***	0.010***
Long-Short	46.40 %	7.66 %	0.83	−6.59 %	(3.33)	(3.71)

This table reports the performance metrics for the 20 fund portfolios constructed using a machine learning combination with three methods, including Random Forest, XGBoost, and LightGBM. The metrics include total return, annualized return, Sharpe ratio, annualized return over the fund market index, and alphas relative to the CH3 and CH4 factor models, with corresponding t-statistics in parentheses. Portfolio returns are net of all costs. Statistical significance is denoted by one, two, and three asterisks at the 10 %, 5 %, and 1 % levels, respectively. The table also includes the long-short (0–19) portfolio, representing the performance difference between the highest and lowest quantile portfolios.

Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Binru Zhao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Acknowledgments

We are grateful to Dimitrios Gounopoulos, David Newton, Yangyi Liu, Weihao Han, Yan Zhao, Qin Gu, Man Wang, Farshad Saadatmand, and participants of the 2022 University of Bath Research Rocket for stimulating discussions and helpful comments on earlier drafts of the paper. Sinolink Securities is a company that provides investment banking and securities brokerage services in China, which may or may not apply similar investment techniques or methods of analysis as described herein. The views expressed here are those of the authors and not necessarily those of Sinolink. The usual caveat applies. Corresponding author: Binru Zhao

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pacfin.2025.102935.

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